OpenEdge Bridge — Client-Side Setup Guide

Version: 1.3
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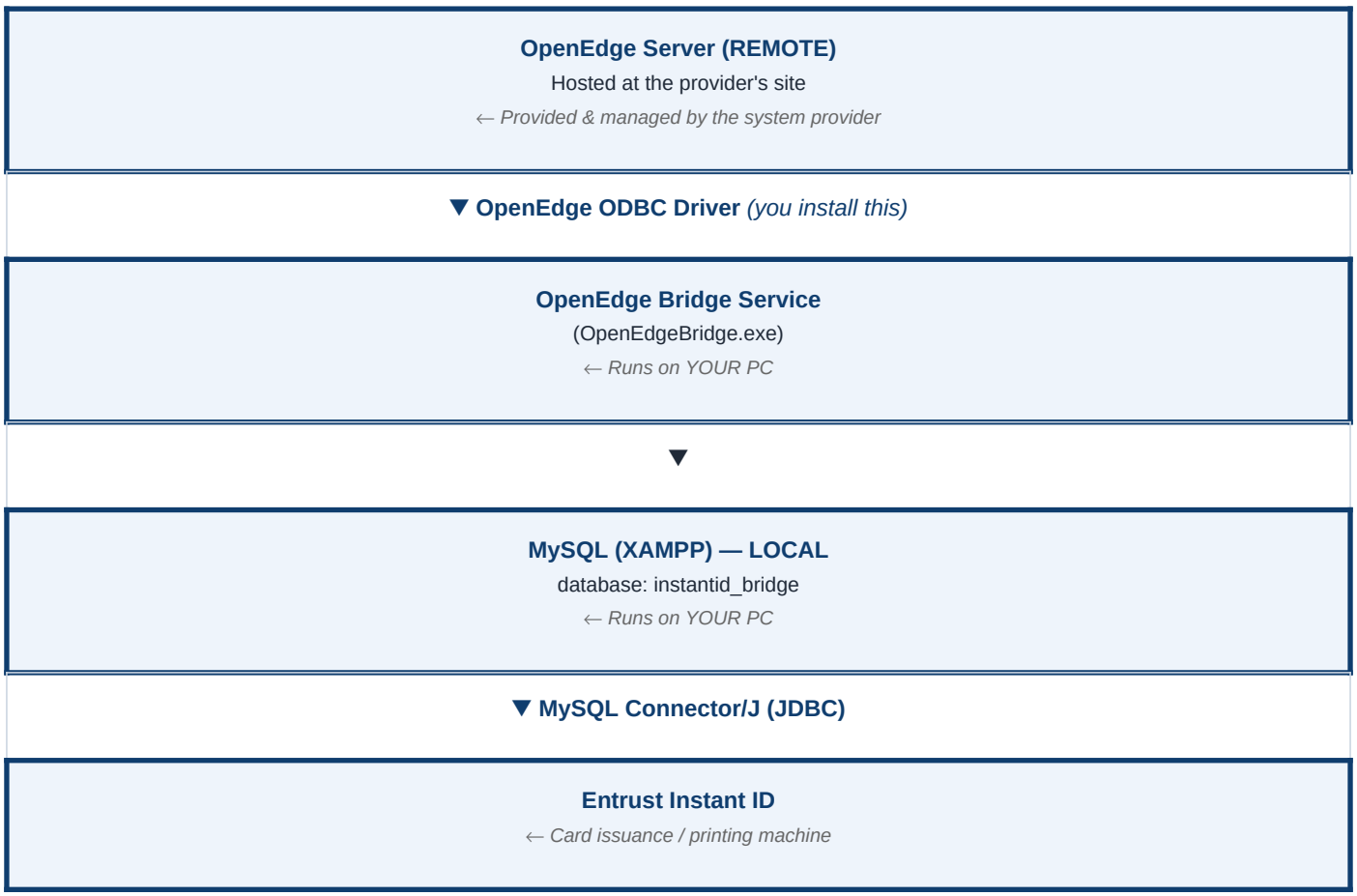
Audience: Client IT / personnel deploying the ID-printing system on their Windows PC.

What This Guide Covers

This guide walks you through setting up everything **on your PC** so that ID-card data flows from the **central OpenEdge server** (managed by the system provider) into a **local MySQL database** (run via XAMPP), which is then read by the **Entrust Instant ID** card-printing machine.



In short, you need to install 4 things on your PC:

#	What	Why
1	**OpenEdge ODBC Driver**	Lets your PC talk to the remote OpenEdge server
2	**XAMPP** (MySQL)	Local database that Entrust reads from

3	**OpenEdgeBridge** (published folder)	The service that copies data from OpenEdge → MySQL
4	**MySQL Connector/J** (`.jar`)	Lets Entrust connect to MySQL

■■ **Important:** The OpenEdge server itself is **remote** — you do **not** install or configure OpenEdge on your PC. You only install the **driver** that connects to it.

Before You Start — Information You Need From the Provider

Ask the system provider for the following details and keep them handy:

Item	Example	Where Used
OpenEdge server **Host** (IP or name)	`10.0.0.25`	ODBC DSN setup (Step 2)
OpenEdge **Port**	`20940`	ODBC DSN setup (Step 2)
OpenEdge **Database name**	`instantID`	ODBC DSN setup (Step 2)
OpenEdge **User ID / Password**	`admin` / `******`	ODBC DSN setup (Step 2)
OpenEdge **DSN name** to use	`OpenEdgeInstantID`	`appsettings.json` (Step 7)
MySQL **password** you want to set	—	XAMPP/phpMyAdmin setup (Step 4)
The **`publish/` folder** of the bridge	—	Deployment (Step 7)

Step 1 — Install the OpenEdge ODBC Driver

1. Get the installer file: `PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe` (provided by the system provider).
2. **Right-click → Run as Administrator** and follow the installer defaults.
3. When finished, verify it is installed by opening **ODBC Data Source Administrator** (64-bit):
4. Press `Win + R`, type `odbcad32.exe`, press **Enter**.
5. You should see **Progress DataDirect OpenEdge SQL Server** (or similar) in the list of available drivers.
6. **Restart your PC** if the driver was just installed (recommended so the driver registers properly).

■ **Done when:** `odbcad32.exe` shows the Progress DataDirect driver in the *Drivers* tab.

Step 2 — Create the ODBC System DSN (Connection to Remote OpenEdge)

A **DSN** (Data Source Name) is a saved "connection profile" that the bridge uses to reach the remote OpenEdge server.

1. Press `Win + R`, type `odbcad32.exe`, press **Enter**.
2. Go to the **System DSN** tab → click **Add...**.
3. Select **Progress DataDirect OpenEdge SQL Server** → click **Finish**.
4. Fill in the details (from the *Before You Start* table):

Field	Value
Data Source Name	`OpenEdgeInstantID` (use exactly this name)
Description	`OpenEdge Instant ID Database`
Host Name	*(provided by provider — e.g., `10.0.0.25`)*
Port Number	*(provided by provider — e.g., `20940`)*
Database Name	*(provided by provider — e.g., `instantID`)*
User ID	*(provided by provider — e.g., `admin`)*
Password	*(provided by provider)*

1. Click **Test Connect** — you should see a success message. If it fails, check the host/port/user/password with the provider.
2. Click **OK** to save the DSN.

■ **Done when: Test Connect** succeeds, and `OpenEdgeInstantID` appears under the **System DSN** tab.

■■ **Permissions on the remote OpenEdge server:** The user you entered in the DSN must be able to read and update the `pub."id-print"` table. If your login user is **not** `admin`, ask the system provider (or run yourself from a DBA account) to execute this on the OpenEdge server, replacing `your_login_user` with your actual DSN user:

```
sql GRANT SELECT, INSERT, UPDATE, DELETE ON pub."id-print" TO your_login_user;
```

Without this permission, the bridge will fail with an access-denied error when it starts.

Step 3 — Install XAMPP (Local MySQL)

1. Download XAMPP from **https://www.apachefriends.org/** (latest version).
2. Run the installer and accept the defaults. Install to the default location (e.g., `C:\xampp`).
3. Open the **XAMPP Control Panel** (from Start Menu or desktop shortcut).
4. Click **Start** next to **MySQL** — the row should turn green.
5. Click **Admin** next to MySQL — this opens **phpMyAdmin** in your browser (http://localhost/phpmyadmin).
6. **Set a password for the MySQL `root` user** (recommended for security):
7. In phpMyAdmin, click the **User accounts** tab.

8. Click **Edit privileges** on `root@localhost` → **Change password** → set a password you will remember.
9. Write this password down — you will need it in Step 7.
10. Keep MySQL **running** (do not close the XAMPP Control Panel, or minimize it).

> ■ **Tip:** For the machine to work after a reboot, add XAMPP's MySQL to auto-start, or simply start MySQL each time the PC boots.

> ■ **Done when:** MySQL shows a green **Running** status in the XAMPP Control Panel and phpMyAdmin opens.

Step 4 — Create the Local Database & Table in MySQL

1. Open **phpMyAdmin** (from XAMPP Control Panel → **Admin** next to MySQL).
2. Click the **SQL** tab and paste the following:

```
CREATE DATABASE IF NOT EXISTS instantid_bridge CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE instantid_bridge;
```

1. Click **Go**.
2. Still in the **SQL** tab (make sure `instantid_bridge` is selected in the left panel), paste and run:

```
CREATE TABLE IF NOT EXISTS id_print ( id INT NOT NULL AUTO_INCREMENT, batch_no INT NULL,
batch_date DATE NULL, app_no INT NULL, dob DATE NULL, cp_no VARCHAR(15) NULL, fullname VARCHAR(65)
NULL, hcp_fullname VARCHAR(65) NULL, address VARCHAR(100) NULL, hcp_cpno VARCHAR(15) NULL, lname
VARCHAR(55) NULL, fname VARCHAR(55) NULL, mname VARCHAR(55) NULL, spa_docno VARCHAR(25) NULL,
page_no VARCHAR(20) NULL, book_no VARCHAR(20) NULL, series_no VARCHAR(20) NULL, spa_atty
VARCHAR(65) NULL, spa_dt_issue DATE NULL, printed VARCHAR(3) NULL COMMENT 'yes / no (matches
OpenEdge LOGICAL FORMAT)', printed_dt DATE NULL, printed_tm VARCHAR(35) NULL, tran_type INT NULL
COMMENT '1 = for RTC, 2 = senior', image_data LONGBLOB NULL, id_code INT NULL, sfx VARCHAR(15)
NULL, photo LONGBLOB NULL, gender INT NULL COMMENT '1 = Female, 2 = Male', date_issued DATE NULL,
id_no VARCHAR(25) NULL, year_applied INT NULL, control_no INT NULL, signature_data LONGBLOB NULL
COMMENT 'Signature image; written by Entrust Instant ID (MySQL only)', qr_data LONGBLOB NULL
COMMENT 'QR code image; written by Entrust Instant ID (MySQL only)', print_count INT NULL COMMENT
'Incremented by Instant ID; triggers printed=yes back-sync', PRIMARY KEY (id), UNIQUE INDEX
idx_app_no (app_no), INDEX idx_batch_no (batch_no), INDEX idx_print_sync (printed, print_count,
app_no) ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

1. Click **Go**.
2. Verify: in the left panel, expand `instantid_bridge` → you should see the `id_print` table.

> ■ **Updating an existing installation (already have `id_print`)?** Run this instead of the CREATE TABLE above, then click **Go**:

```
sql ALTER TABLE id_print MODIFY COLUMN address VARCHAR(100) NULL, CHANGE COLUMN id_type tran_type
INT NULL COMMENT '1 = for RTC, 2 = senior', ADD COLUMN sfx VARCHAR(15) NULL AFTER id_code, ADD
COLUMN photo LONGBLOB NULL AFTER sfx, ADD COLUMN gender INT NULL COMMENT '1 = Female, 2 = Male'
AFTER photo, ADD COLUMN date_issued DATE NULL AFTER gender, MODIFY COLUMN id_no VARCHAR(25) NULL
```

```
AFTER gender, ADD COLUMN year_applied INT NULL AFTER id_no, ADD COLUMN control_no INT NULL AFTER year_applied, ADD COLUMN signature_data LONGBLOB NULL AFTER control_no, ADD COLUMN qr_data LONGBLOB NULL AFTER signature_data, ADD INDEX idx_batch_no (batch_no), ADD INDEX idx_print_sync (printed, print_count, app_no);
```

(If you already had `signature_data` and `qr_data`, just run the `CHANGE COLUMN id_type` line and the first seven `ADD COLUMN` lines.)

■ **Done when:** `instantid_bridge` database and `id_print` table appear in phpMyAdmin.

Step 5 — Configure Entrust Instant ID (MySQL Connector/J)

1. Copy the file **`mysql-connector-j-8.0.33.jar`** (provided by the system provider) into Entrust Instant ID's `lib/` folder.
2. In **Entrust Instant ID**, configure the database connection to point to your **local** MySQL:

Setting	Value
Host	`localhost`
Port	`3306`
Database	`instantid_bridge`
User	`root` (or a dedicated user you created)
Password	*(the MySQL password you set in Step 3)*
Table	`id_print`

1. Save the settings and restart Entrust Instant ID.

■■ Entrust connects to MySQL **on your PC** (`localhost`) — not to OpenEdge. The bridge is what talks to OpenEdge.

■ **Done when:** Entrust Instant ID can open/query the `id_print` table without errors.

Step 6 — Deploy the Bridge Service

1. Copy the entire **`publish/`** folder (provided by the system provider) to your PC, e.g. to **`C:\OpenEdgeBridge\`**.
2. The folder must contain at least: `OpenEdgeBridge.exe`, `appsettings.json`, and the `runtimes/` subfolder.
3. **Edit `appsettings.json`** (right-click → Open with Notepad) and fill in your real values:

```
{ "ConnectionStrings": { "OpenEdge":
"DSN=OpenEdgeInstantID;Uid=admin;Pwd=YOUR_OPENEDGE_PASSWORD;DefaultIsolationLevel=1;",
"OpenEdgeRead":
"DSN=OpenEdgeInstantID;Uid=admin;Pwd=YOUR_OPENEDGE_PASSWORD;DefaultIsolationLevel=1;", "MySql":
"Server=127.0.0.1;Database=instantid_bridge;Uid=root;Pwd=YOUR_MYSQL_PASSWORD;" }, "Sync": {
"IntervalSeconds": 10, "FullSyncIntervalMinutes": 0 } }
```

Setting	What to put
`OpenEdge` / `OpenEdgeRead`	`DSN` must be **exactly** the DSN name from Step 2 (`OpenEdgeInstantID`); `Uid`/`Pwd` are the OpenEdge credentials from the provider
`MySql`	`Server=127.0.0.1` (your local XAMPP); `Uid=root`; `Pwd` = the MySQL password you set in Step 3
`Sync:IntervalSeconds`	How often the bridge checks for new data (10 = every 10 seconds; 10–30 is recommended)
`Sync:FullSyncIntervalMinutes`	Keep at **`0`**. The bridge then copies everything once when it first runs, and afterwards only picks up **new batches** — every 10 seconds, forever, even after reboots. Do not change this unless the provider tells you to

1. Save the file.

> ■■ **Do not** rename or move `appsettings.json` — it must sit in the same folder as `OpenEdgeBridge.exe`.

> ■ **Done when:** `C:\OpenEdgeBridge\` contains `OpenEdgeBridge.exe` and an edited `appsettings.json`.

Test the bridge manually in PowerShell

Before installing the Windows service, run the executable manually to verify the configuration and database connections:

```
Set-Location C:\OpenEdgeBridge .\OpenEdgeBridge.exe
```

Keep the PowerShell window open while testing. The bridge runs in the foreground and displays connection or sync errors in the console. Confirm that rows appear in `instantid_bridge.id_print`, then press **Ctrl+C** to stop the bridge.

> ■■ Stop the manual test with **Ctrl+C** before starting the Windows service, otherwise both processes may run at the same time.

Step 7 — Install the Bridge as a Windows Service

1. Open **PowerShell as Administrator** (right-click Start → *Windows PowerShell (Admin)* / *Terminal (Admin)*).
2. Run (adjust the path if you copied the folder elsewhere):

```
New-Service -Name "OpenEdgeBridge" ` -BinaryPathName "C:\OpenEdgeBridge\OpenEdgeBridge.exe" ` -DisplayName "OpenEdge Bridge Service" ` -Description "Syncs id-print data between OpenEdge and MySQL for Entrust Instant ID" ` -StartupType Automatic Start-Service OpenEdgeBridge
```

1. Verify it is running:

```
Get-Service OpenEdgeBridge # Status should be: Running
```

1. **Confirm data is syncing:** open phpMyAdmin → `instantid_bridge` → `id_print` → **Browse**. After a few seconds, rows should appear that match the records in OpenEdge.

■ **Done when:** Service status = **Running**, and rows appear in `id_print` automatically.

How the Sync Works (Plain English)

This is what the bridge does behind the scenes — you normally never have to think about it:

1. **First time it runs**, it copies **everything** from the OpenEdge `id-print` table into your local MySQL (`id_print`). This can take a few minutes if there are many records — that is normal.
2. **After that**, every 10 seconds it only asks OpenEdge for **new batches** (new groups of applications) and copies just those. Existing rows are not re-copied.
3. The bridge remembers this in a small bookkeeping table (`sync_state`) inside MySQL — so **restarting the PC or the service never causes another full copy**. It picks up right where it left off.
4. Separately, it checks every 10 seconds whether Entrust Instant ID has printed anything (`print_count`), and marks those records as `printed = yes` back in OpenEdge.

■ **Expect this once:** if you ever upgrade the bridge from an older version, the first start after the upgrade does one full copy again (to create the bookkeeping table), then returns to new-batches-only.

Everyday Operations

Start / Stop / Restart the service

```
Start-Service OpenEdgeBridge Stop-Service OpenEdgeBridge Restart-Service OpenEdgeBridge # use this after changing appsettings.json Get-Service OpenEdgeBridge # check status
```

Daily startup order (after a PC reboot)

1. Open **XAMPP Control Panel** → click **Start** next to **MySQL**.
2. The **OpenEdge Bridge** service starts automatically (StartupType = Automatic). If not, run `Start-Service OpenEdgeBridge`.
3. Check that `id_print` in phpMyAdmin is receiving rows.

Checking the logs

- Log files: `C:\OpenEdgeBridge\Logs\` (one dated file per day)
- Windows Event Viewer: *Windows Logs* → *Application*, source: `OpenEdgeBridge`

Troubleshooting (Common Issues)

Problem	Likely Cause	Fix
Service starts then stops immediately	Wrong DSN name or wrong OpenEdge credentials	Open `odbcad32.exe` → select `OpenEdgeInstantID` → **Configure** → **Test Connect**. Then `Restart-Service OpenEdgeBridge`
"Driver not found" / no DSN listed	Wrong bitness (32-bit vs 64-bit)	Always use **`odbcad32.exe`** (64-bit) — NOT the 32-bit one in `C:\Windows\SysWOW64`
No rows in `id_print`	MySQL not running, or bridge not running	Check XAMPP Control Panel → MySQL must be **green**; check `Get-Service OpenEdgeBridge`
"MySQL connection string not found"	`appsettings.json` missing or not beside the `.exe`	Confirm `appsettings.json` is in `C:\OpenEdgeBridge\` and has valid JSON
Duplicate rows in MySQL	Missing UNIQUE index on `app_no`	Run: `ALTER TABLE id_print ADD UNIQUE INDEX idx_app_no (app_no);`
Entrust can't connect to MySQL	MySQL not started, or wrong credentials in Entrust	Check XAMPP (MySQL green); re-check Entrust host `localhost`, port `3306`, database `instantid_bridge`
`printed` never becomes `yes` in OpenEdge	Entrust isn't incrementing `print_count`	Check Entrust is configured to the `id_print` table and is incrementing `print_count` after printing
Bridge keeps copying all records on every start	MySQL unreachable at startup, or `sync_state` table missing	The bridge falls back to a full copy when it can't read its state. Check XAMPP MySQL is **green** and that a `sync_state` table exists in `instantid_bridge` (the bridge creates it automatically after the first successful run)
Bridge can't reach remote OpenEdge	Network/firewall blocking, or wrong host/port	Verify you can ping the host and the port is reachable; confirm details with the provider

Checklist for Your PC

- [ ] OpenEdge ODBC Driver installed (64-bit)
- [ ] System DSN `OpenEdgeInstantID` created and **Test Connect** succeeds
- [ ] XAMPP installed, MySQL **running**, `root` password set
- [ ] `instantid_bridge` database created
- [ ] `id_print` table created (with `tran_type`, `sfx`, `photo`, `gender`, `date_issued`, `id_no`, `year_applied`, `control_no`, `signature_data`, `qr_data`, `print_count`, and UNIQUE `app_no`)
- [ ] `sync_state` table exists in MySQL (auto-created by the bridge after the first run — verify in phpMyAdmin)
- [ ] `mysql-connector-j-8.0.33.jar` placed in Entrust's `lib/` folder
- [ ] Entrust configured → `localhost:3306` / `instantid_bridge` / `id_print`
- [ ] `publish/` folder copied to `C:\OpenEdgeBridge\`
- [ ] `appsettings.json` edited with correct DSN name, OpenEdge user/pass, MySQL user/pass
- [ ] Windows Service installed and **Running**
- [ ] Rows visible in `id_print` (phpMyAdmin) confirming sync works
- [ ] `C:\OpenEdgeBridge\Logs\` has today's log file with no errors

Contact / Escalation

If any step fails after following this guide:

1. Check the **Logs** folder (`C:\OpenEdgeBridge\Logs\`) for error messages.
2. Send the log file and the output of `Get-Service OpenEdgeBridge` to the system provider.
3. Provide the result of the ODBC **Test Connect** (screenshot) — this isolates whether the issue is the driver/network or the bridge itself.